

MathCamp 2025 — Slide Deck Overview

Table of contents drafted from last year's slide-based sessions

Draft prepared for MathCamp 2026 planning

2026-07-23

Chapter 1

Purpose

This document is a working table of contents (down to the third level) for the five slide decks used in the 2025 MathCamp session (**Resources/Slide Decks 2025/**). It's meant to give a quick overview of what was covered last year, to inform how we scope and sequence the new Quarto-book-based, interactive version of the material for 2026.

Chapter 2

Lecture 1 — Notation, Functions, and Limits

SlideDeck1_081825.pdf, Aug 18, 2025

2.1 Introduction & Logistics

- Introductions
- Math camp goals, logistics, and schedule
- What are quantitative methods? Quantitative vs. qualitative methods

2.2 Variables & Measures

- Variables and constants
- Sets and sample spaces
- Unions, intersections, subsets (set notation)
- Levels of measurement, and why it matters

2.3 Rules of Algebra

- Basic properties of arithmetic
- Order of operations
- Inequalities
- Exponents, logarithms, and roots

2.4 Functions

- Functions and their characteristics

- Function composition
- Graph examples: linear, quadratic, exponential functions
- Functions and theorizing; Anscombe's quartet

2.5 Solving Equations

- Special products and factoring
- Quadratics and special products
- Quadratic formula, checking work graphically

2.6 Series, Sequences, and Limits

- Sequences and series
 - Limits (including piecewise function examples)
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Chapter 3

Lecture 2 — Probability

SlideDeck2_081925.pdf, Aug 19, 2025

3.1 Introduction

- Agenda
- Frequentist vs. Bayesian framing

3.2 Bayesian Probability

- Definitions and notation
- Relationships between events
- Bayes' theorem (notation, derivation, examples)
- The Monty Hall problem
- Odds and odds ratios

3.3 Frequentist Statistics

- Frequentist methods and uncertainty
- Confidence interval demonstration in R

3.4 Combinations and Permutations

- Review of factorials
- Combinations (why the formulas work, worked examples)
- Calculating probabilities; the birthday problem

3.5 Distributions

- Random variables: discrete vs. continuous
 - Sample distributions and contingency tables
 - Probability distributions: PMF, PDF, CDF
 - Normal/Gaussian distribution and other distribution types
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Chapter 4

Lecture 3 — Calculus I: Differentiation & Optimization

SlideDeck3_082025.pdf, Aug 20, 2025

4.1 Differentiation

- Types of change and derivatives
- Discrete change (secant lines) vs. instantaneous change (tangent lines)
- Rules of differentiation: power, product, quotient, and chain rules
- Worked examples and practice

4.2 Optimization

- What is optimization?
 - Extreme value theorem
 - Derivatives and the shape of a function
 - Inflection points vs. extrema
 - Worked optimization example
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Chapter 5

Lecture 4 — Calculus II: Integration & Multivariate Calculus

SlideDeck4_082125.pdf, Aug 21, 2025

5.1 Integration

- Area under the curve
- Integrals and the fundamental theorem of calculus
- Reversing the rules of differentiation (“don’t forget the C!”)
- Rules of integration, worked examples
- Integration by substitution

5.2 Multivariate Calculus

- Partial derivatives
 - Gradients and mixed partial derivatives
 - The Hessian matrix
 - Application: deriving Ordinary Least Squares (OLS)
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Chapter 6

Lecture 5 — Linear Algebra

SlideDeck5_082225.pdf, Aug 22, 2025

6.1 Equations & Systems of Equations

- Equations and unknowns
- Solving by substitution and elimination
- More complex systems of equations

6.2 Vectors

- Vectors and scalars
- Vector addition/subtraction, scalar multiplication
- Dot product

6.3 Matrices

- Important types of matrices
- Matrix transposition
- Matrix addition, subtraction, and scalar multiplication
- Matrix multiplication

6.4 Determinants

- Solving for a determinant
- Worked full example

6.5 Inverse Matrices

- Deriving the inverse of a 2×2 matrix
- Inverting 2×2 and 3×3 matrices (minors, cofactor matrix, transpose)

6.6 Matrix & Vector Properties

- Linear independence
- Matrix rank

6.7 Systems of Equations as Matrices

- Solving systems via matrix inversion, worked example
 - Overdetermined systems of equations
 - Best-fit line via matrices (connects back to OLS from Lecture 4)
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Chapter 7

Notes for the 2026 redesign

- Topic flow across the week runs: algebra/functions/limits $\rightarrow$ probability $\rightarrow$ differential calculus $\rightarrow$ integral/multivariate calculus $\rightarrow$ linear algebra, with OLS introduced twice (as a multivariate-calculus application in Lecture 4, and derived via matrices in Lecture 5).
- Candidate spots for interactive visualizations: set operations (Venn diagrams), function graphing, limits, the Monty Hall problem, distribution shapes (PMF/PDF/CDF), tangent/secant lines, optimization surfaces, gradients/Hessian, and matrix operations (esp. determinants and inverses).